

INTEGRATED CLIMATE-INTELLIGENT SALT PRODUCTION ECOSYSTEM: A MULTI-MODAL AI FRAMEWORK FOR SUSTAINABLE YIELD OPTIMIZATION IN TROPICAL COASTAL ENVIRONMENTS

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Deployment Report

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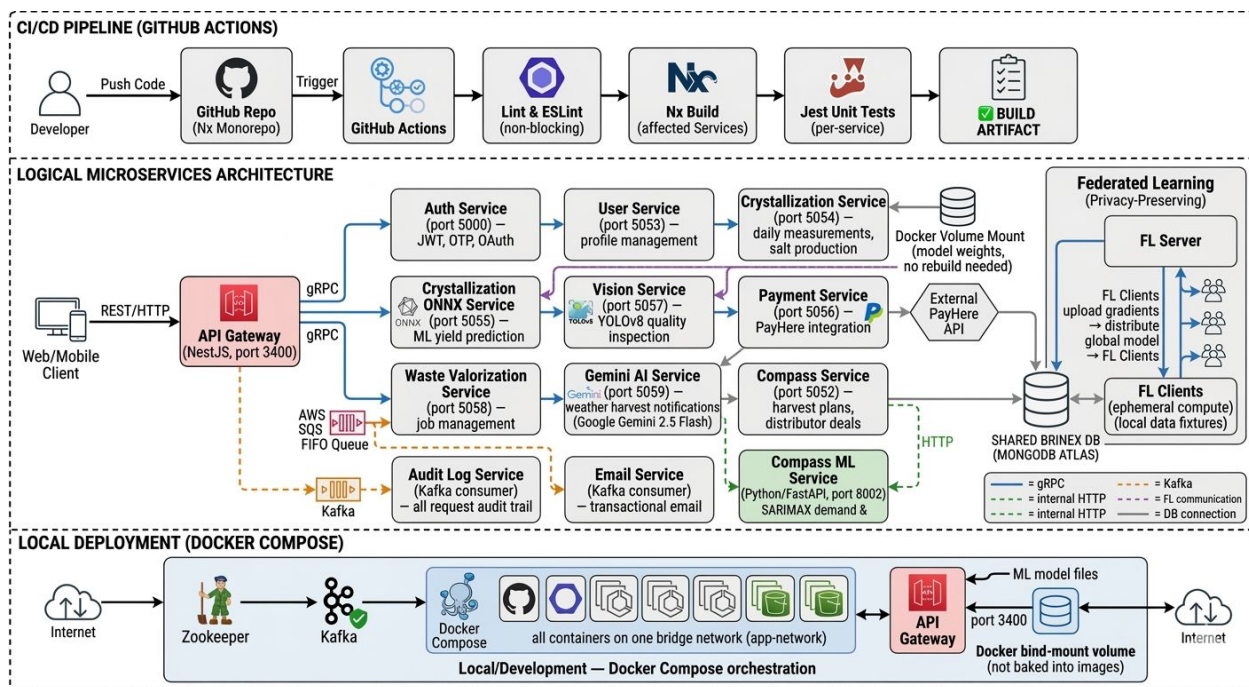
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1. Executive Summary

This report details the deployment architecture, configuration, and validation protocols for the BrineX is a cloud-native, production-grade backend platform comprising 13 independent microservices - 11 in TypeScript/NestJS and 2 in Python/FastAPI - orchestrated within an Nx monorepo. It exposes a unified REST API (API Gateway) that fans out to domain services over gRPC (synchronous) and Apache Kafka (asynchronous). Persistence is via MongoDB Atlas. Key services include ONNX-based salt crystallization prediction, YOLOv8 computer vision quality inspection, SARIMAX demand/price forecasting, Gemini 2.5 Flash AI weather notifications, AWS SQS waste valorization job processing, and PayHere payment processing. All services are containerised with Docker multi-stage builds and orchestrated via Docker Compose. Next.js frontend



2. Software Specifications & Architecture

The system is built using a polyglot microservices approach, managed within an **Nx Monorepo** to ensure consistent tooling and shared libraries across services.

2.1 Technology Stack

Layer	Technology	Version
Runtime	Node.js	20 (LTS)
Language (Services)	TypeScript	~5.9.2
Language (ML Service)	Python/ PyTorch	3.11

Backend Framework	NestJS	^11.0.0
Frontend Application	Next.js / React	Next 16.0.7 / React 19.2
ML HTTP Framework	FastAPI + Uvicorn	0.111.0 / 0.29.0
Monorepo Tooling	Nx	21.5.2
Database	MongoDB Atlas (Mongoose ODM)	^8.18.1
Message Broker	Apache Kafka (Confluent) + Zookeeper	7.5.0
Inter-Service RPC	gRPC over Protocol Buffers (proto3)	@grpc/grpc-js ^1.13.4
ML Inference (Node)	ONNX Runtime Node	^1.24.1
Containerisation	Docker (multi-stage)	—
CI/CD	GitHub Actions	—
Authentication	JWT + Passport	^11.0.0 / ^4.0.1
Payment Gateway	PayHere	REST webhook
AI / LLM	Google Gemini 2.5 Flash	—
Push Notifications	Expo Server SDK	^4.0.0
Task Queueing	AWS SQS	Standard Queue with DLQ

3. Tooling & Dependency Analysis

3.1 Core Backend & Frontend Libraries

- **Backend:** @nestjs/microservices for gRPC/Kafka communication, mongoose for data modeling, and onnx runtime-node for serving optimized ML models within the Node environment.
- **Frontend:** zustand for state management, framer-motion for UI/UX fluidity, and recharts for data visualization of waste metrics

3.2 Machine Learning Environment

The ML services utilize a specialized stack to balance performance and deployment size:

- The Python-based compass-ml-service runs as a FastAPI microservice with: fastapi 0.111.0, uvicorn 0.29.0, pydantic 2.7.1, statsmodels 0.14.2 (SARIMAX forecasting), pandas 2.2.2, numpy 1.26.4, scipy 1.13.0, pymongo 4.7.2.
- Model artefacts: price_sarimax.pkl, price_sarimax_meta.json, price_sarimax_history.pkl (compass-ml-service);
- best.onnx (YOLOv8, vision-service);
- crystallization ONNX models (crystallization-onnx-service).

- Waste Prediction Library: Custom internal package managing torch, xgboost, and lightgbm models.
- **All model files are injected via Docker volume mounts, not baked into images, enabling model updates without container rebuilds.**

4. Deployment Processes & Workflows

4.1 Development & Local Orchestration

Local development is standardized using **Docker Compose** to ensure "it works on my machine" parity.

1. **Environment Initialization:** Secrets are mapped from .env (derived from .env.example).
2. **Service Spin-up:** docker compose up --build initializes the infrastructure (Kafka, Zookeeper, Mongo) alongside the services.
3. **Frontend Live-Reload:** npm run dev in the frontend directory for rapid UI iteration.

4.2 CI/CD Pipeline (GitHub Actions)

There are 10 independent per-service workflow files, each following a sequential 2-job pipeline:

Job 1 - test: Runs ESLint (non-blocking, continue-on-error: true) followed by Jest unit tests per service. Triggered on push or pull_request to master / develop using path-filtered globs so a workflow only triggers when its relevant service, shared packages, or proto files change.

Job 2 - build: Runs only after test passes. Executes npx nx build <service> --skip-nx-cache and uploads the dist artifact (7-day retention).

Planned CI/CD additions for AWS deployment:

- Docker image build and push to **Amazon ECR** after successful nx build
- Container vulnerability scanning via **Trivy** as a gate before ECR push
- Automated deployment trigger to **AWS ECS (Fargate)** on successful image push to ECR

5. Cloud Infrastructure & Configuration

5.1 Planned AWS Production Architecture

The production deployment targets **AWS ECS (Fargate)** with the following planned infrastructure:

Networking: All backend microservices will be deployed into **Private Subnets** within a dedicated **VPC**. Only the API Gateway will be publicly reachable via an **Application Load Balancer (ALB)** in a Public Subnet. All inter-service gRPC communication will remain on the private network using container DNS resolution within the ECS cluster, mirroring the current Docker Compose app-network behavior.

Secret Management: All environment variable secrets (MongoDB URI, JWT secret, Gemini API key, PayHere credentials, AWS keys, SMTP credentials) currently injected via .env will be migrated to **AWS Secrets Manager**, with runtime injection into ECS task definitions via secrets configuration. No credentials will be stored in the repository or baked into container images.

IAM Strategy: Each ECS task will be assigned a dedicated **IAM Role** with least-privilege permissions - for example, the waste-valorization-service task role will have permission only to send messages to its designated SQS FIFO queue, and no other service will have SQS access.

Container Registry: All 13 service images will be built via multi-stage Dockerfiles and stored in **Amazon ECR**, one repository per service.

5.2 Storage & Messaging Policy

MongoDB Atlas:

The existing single shared MongoDB Atlas cluster (project: brinex) will be retained in production. Atlas provides managed backups, auto-scaling, and multi-region replication without requiring self-hosted MongoDB on AWS. The MONGO_URI will be injected at runtime via AWS Secrets Manager.

Apache Kafka:

The current single-broker Confluent Kafka setup (replication factor 1) is a development configuration and is not suitable for production HA. For production, this will be migrated to Amazon MSK (Managed Streaming for Apache Kafka) with a minimum of 3 brokers and replication factor 3 to ensure fault tolerance.

AWS SQS (FIFO Queue):

Already production-ready. The waste-valorization-service uses an SQS FIFO queue ({ENVIRONMENT}_waste_prediction_requests.fifo) for exactly-once, ordered job processing. A Dead Letter Queue (DLQ) will be configured with a visibility timeout of 300s to capture and retain failed job processing attempts for analysis.

AWS S3:

ML model artefacts (price_sarimax.pkl, best.onnx, crystallization ONNX models, FL global model weights) will be stored in a versioned S3 bucket, replacing the current Docker bind-mount approach. ECS tasks will pull the latest model version at container startup, enabling model updates without a full container rebuild or redeployment.

Federated Learning Infrastructure:

The FL Server will be deployed as a dedicated ECS Fargate task. FL Clients will use ephemeral compute (AWS Lambda or short-lived Fargate tasks) to train on local data fixtures and upload gradient updates. Model aggregation artefacts and global model versions will be stored in the versioned S3 bucket and tracked in MongoDB Atlas.

6. Federated Learning (FL) Architecture

The FL implementation is designed for edge-compatibility.

- **Server Side:** Coordinates aggregation rounds and model distribution.
- **Client Side:** Uses ephemeral compute to train on local fixtures, ensuring data privacy before uploading weight updates.
- **Security:** Model updates are signed and transmitted over encrypted channels to prevent poisoning attacks.

7. Testing & Quality Assurance

7.1 Automated Test Execution

Bash

Workspace-wide Unit Tests

npm run test:unit

Integration Tests (requires infrastructure mocks)

npm run test:integration

7.2 Core Research Test Cases

ID	Test Case	Description	Steps to execute	Expected Visual/UI Result	Priority	1st Round	2nd Round
FE-AUTH-001	Landing Page Layout	Verifies the initial entry point	Open the application URL.	The "Intelligent Salt" landing page appears with BrineX logo and three distinct buttons: "Login as Landowner / Distributor", "Login as Laboratory", and "Admin Login".	Critical	Pass	
FE-AUTH-002	Landowner/Distributor Tab Toggle	Validates the segmented control switch on the Sign In card	1. Click "Login as Landowner / Distributor" from landing page. 2. Toggle between "Landowner" and "Distributor" tabs.	The active tab background turns white with a shadow, while the inactive tab remains gray. The "Phone Number" label remains consistent.	High	Pass	
FE-AUTH-003	Laboratory Login Link Navigation	Verifies the cross-link navigation between login types.	Click the "Login as Laboratory" blue text link at the bottom of the Sign In card.	The UI should transition from the Phone OTP card to a different login card (typically email-based).	High	Pass	
FE-AUTH-004	Phone Input UI & Masking	Checks the visual formatting of the phone input field.	1. Observe the phone input field. 2. Type "771234567".	The "+94" prefix is locked in a separate gray box. The input field displays a phone icon and a placeholder "7X XXX XXXX".	Medium	Pass	
FE-AUTH-005	Empty Field Validation UI	Verifies that the UI alerts the user when clicking without input.	Click "Send OTP" button without entering any numbers.	A red validation message appears below the input field: "Phone number is required" or the input border turns red.	Critical	Pass	

FE-AUTH-006	"Send OTP" Button State	Validates visual feedback during the click action.	Click the "Send OTP" button.	The button should show a click animation or a brief loading spinner (ssl.png), indicating the action is being processed.	Medium	Fail	Pass
FE-AUTH-007	Admin Login Page Visuals	Verifies the Super Admin card layout	Click the "Admin Login" small link from the landing page.	A "Super Admin Login" card appears with Email and Password icons inside the input fields and a dark "Sign In" button.	High	Pass	
FE-AUTH-008	Password Visibility Toggle	Checks if the user can view/hide admin passwords.	1. Enter text in the Password field. 2. Look for/click a visibility icon (if present).	The password characters should be masked by dots by default.	Medium	Pass	
FE-AUTH-	Error Banner Clearance	Ensures the UI resets error states visually.	1. Trigger an error (e.g., wrong input). 2. Start typing a new value.	The red error banner or red border should disappear as soon as the user starts typing fresh data.	Medium	Pass	
FE-AUTH-010	Responsive Layout Check	Verifies UI integrity on smaller screens.	Resize the browser window to a mobile width.	The "Intelligent Salt" text should scale, and the login cards should remain centered and fully visible without horizontal scrolling.	High	Pass	
FE-AUTH-018	Role Selection Display	Verifies that all three registration paths are visually available.	Navigate to the Sign Up page (/auth/signup).	Three distinct role buttons are visible: Landowner, Distributor, and Laboratory.	Critical	Pass	
FE-AUTH-019	Landowner Field Toggle	Ensures the UI displays phone-specific inputs for Landowners.	Click the "Landowner" role button.	The Phone Number input field is displayed; the Email Address field is hidden.	High	Pass	

FE-AUTH-020	Laboratory Field Toggle	Ensures the UI displays email-specific inputs for Laboratories.	Click the "Laboratory" role button.	The Email Address input field is displayed; the Phone Number field is hidden.	High	Pass	
FE-AUTH-021	Phone Sign-up Submission	Validates the visual transition after submitting a valid phone number.	1. Select Landowner. 2. Enter a valid phone number. 3. Click the submit button.	The UI transitions to the OTP (One-Time Password) entry screen.	Critical	Pass	
FE-AUTH-022	Email Sign-up Submission	Validates the visual transition after submitting a valid email.	1. Select Laboratory. 2. Enter a valid email address. 3. Click the submit button.	The UI transitions to the OTP entry screen.	Critical	Fail	Pass
FE-AUTH-023	OTP Verification UI	Checks the interface for entering the 6-digit verification code.	1. Complete a sign-up step to receive an OTP. 2. Enter a 6-digit code.	The code characters are visible as they are typed, and the "Verify" button becomes active.	Critical	Fail	Pass
FE-AUTH-024	Post-Signup Redirection	Verifies that new users are guided to the next setup phase.	Enter the correct OTP for a new registration.	The browser automatically redirects the user to the Onboarding page (/auth/onboarding).	Critical	Pass	
FE-AUTH-025	Duplicate User Error UI	Checks the visual feedback when using an existing account.	Attempt to sign up using a phone or email that is already registered.	A red error message or alert banner appears stating that the user already exists.	High	Pass	

FE-ONB-001	Landowner Form Display	Verifies specific fields for the Landowner profile.	Navigate to a LANDOWNER.	UI displays fields for: NIC Number (12 digits), Address, and Total Salt Beds.	Critical	Pass	
FE-ONB-002	Landowner Valid Submission	Confirms UI transition after providing complete data.	1. Enter 12-digit NIC. 2. Enter address and 5 salt beds. 3. Click "Submit".	UI redirects the user to the Plan Selection page (/auth/plans).	Critical	Pass	
FE-ONB-003	NIC Validation UI	Checks visual feedback for incorrect NIC length.	Enter a 5-digit NIC in the input field.	A validation error message appears: "NIC must be 12 digits".	High	Pass	
FE-ONB-	Required Field UI	Validates that the UI flags missing mandatory data.	Leave the "Address" field empty and attempt to submit.	The empty field is highlighted in red with a "required" validation error.	High	Pass	
FE-ONB-005	Salt Bed Minimum UI	Tests input constraints for salt production data.	Enter "0" or a negative number in "Total Salt Beds".	UI displays a validation error: "minimum 1 bed".	Medium	Pass	
FE-ONB-006	Laboratory Form Display	Verifies specific fields for the Laboratory profile.	Navigate to as a LABORATORY.	UI displays fields for: Laboratory Name, Registration Number, and Address.	Critical	Pass	
FE-ONB-007	Laboratory Valid Submission	Confirms UI transition for Lab users.	Fill all Lab fields and click "Submit".	UI redirects to the Plan Selection page.	Critical	Pass	

FE-ONB-008	Distributor Form Display	Verifies specific fields for the Distributor profile.	Navigate to as a DISTRIBUTOR.	UI displays fields for: Company Name, Registration Number, and Address.	Critical	Pass	
FE-ONB-009	Distributor Valid Submission	Confirms UI transition for Distributor users.	Fill all Distributor fields and click "Submit".	UI redirects to the Plan Selection page.	Critical	Fail	Pass
FE-ONB-010	Admin Bypass Logic	Ensures administrators are not forced through onboarding.	Logged in as an ADMIN.	UI automatically redirects to the Admin Dashboard.	High	Pass	
FE-ONB-011	Duplicate Access Redirect	Prevents onboarded users from re-entering the process.	Visit as a user who is already onboarded.	UI automatically redirects to the plans page or the dashboard.	Medium	Pass	
FE-	Landowner Plan Layout	Verifies correct plans are shown for producer roles.	Navigate to LANDOWNER.	Two plan cards appear: Free (0 LKR) and Pro (1,500 LKR/month).	Critical	Pass	
FE-PLAN-002	Laboratory Plan Layout	Verifies specific plan presentation for Lab roles.	Navigate to LABORATORY.	A single Lab Plan card appears (2,500 LKR/month).	Critical	Pass	
FE-PLAN-003	Billing Toggle UI	Tests the visual switch between monthly and annual rates.	Click the "Monthly/Annual" billing toggle.	Card prices update to annual totals; a "17% savings" badge or label becomes visible.	High	Pass	

FE-PLAN-004	Free Plan Selection	Confirms standard navigation for the entry-level plan.	Click "Get Started" on the Free plan card.	UI transitions immediately to the dashboard after a brief loading state.	High	Pass	
FE-PLAN-005	Checkout Initialization	Validates visual feedback when starting a paid subscription.	Click "Subscribe" on the Pro plan card.	The button enters a loading state, and the UI prepares to hand off to the payment gateway.	Critical	Pass	
FE-PLAN-006	PayHere Gateway UI	Verifies successful loading of the external payment interface.	Wait for the checkout process to initiate.	The PayHere checkout modal or page opens, pre-populated with correct order details (amount, currency).	Critical	Fail	Pass
FE-PLAN-007	Checkout Loading UI	Checks for "Double Click" prevention during payment.	Click the "Subscribe" button and observe.	The button displays a spinner and becomes disabled (unclickable) during the request.	Medium	Pass	
FE-PLAN-	Payment API Error UI	Validates user feedback for failed checkout requests.	Attempt to subscribe (simulating a network or API failure).	A red error banner appears at the top of the card or page with a descriptive message.	High	Pass	
FE-PLAN-009	Trial Status UI	Verifies trial information is visible to eligible users.	Access plans as a user with an active trial.	An indicator showing "X days remaining" on the trial is clearly visible on the UI.	Medium	Pass	
FE-PLAN-010	Annual Math Accuracy	Ensures annual prices reflect the correct calculated totals.	Toggle to annual billing and check amounts.	Pro Plan displays 15,000 LKR/year; Lab Plan displays 25,000 LKR/year.	High	Pass	

FE-PAY-001	Payment Success UI	Verifies the visual confirmation of a completed transaction.	Navigate to payments	A green success icon, the message "Subscription Activated," and a prominent "Continue" button are displayed.	Critical	Pass	
FE-PAY-002	Success Page Data Sync	Confirms the UI reflects the new subscription status immediately.	Observe the page load and check user profile details.	The system calls the backend to refresh user details; restricted features (like AI detection) should now appear available in the navigation.	High	Pass	
FE-PAY-003	Role-Based Routing	Validates that the "Continue" button directs users to the correct module.	Click the "Continue" button as a LANDOWNER or LABORATORY user.	Landowners are redirected to /compass/landowner-dashboard; Laboratory users are redirected to /vision/dashboard/camera.	Critical	Pass	
FE-PAY-004	Payment Cancel UI	Verifies the layout when a user aborts the payment process.	Navigate to /payments/cancel.	A cancellation message is shown alongside "Back to Plans" and "Go Home" buttons.	High	Pass	
FE-PAY-	Return to Plans Flow	Checks if users can easily retry the payment process.	Click the "Back to Plans" button from the cancel page.	The UI redirects correctly back to the /auth/plans selection screen.	Medium	Pass	
FE-CMP-001	Welcome Header Display	Verifies personalized greeting and location context.	Load the Landowner dashboard.	"Good morning, {userName}" displays prominently with the user's current location.	High	Pass	
FE-CMP-002	Weather Widget Integration	Checks for real-time environmental data visualization.	Load the dashboard.	A widget displays temperature, weather condition (e.g., "Sunny"), and a corresponding weather icon.	Medium	Pass	

FE-CMP-003	Active Plan Card (Data)	Validates the presentation of an existing harvest plan.	Load dashboard for a user with a saved plan.	A clickable card shows duration, bed count, and start date.	High	Pass	
FE-CMP-004	Active Plan Card (Empty)	Verifies the Call-to-Action for new users.	Load dashboard for a user with no saved plan.	An empty state appears with a "Create your first harvest plan" button.	High	Pass	
FE-CMP-005	Plan Detail Navigation	Ensures the plan card acts as a functional link.	Click on an active harvest plan card.	The UI transitions to the plan creation or detailed view.	Medium	Pass	
FE-CMP-006	Harvest Readiness Metrics	Checks the rendering of salt production data cards.	Load the dashboard.	Metric cards for harvest readiness are visible (displaying mock data).	Medium	Pass	
FE-CMP-007	Main Planner Navigation	Verifies the navigation link to the planner tool.	Click the "Planner" button in the navigation bar.	The UI navigates to the start of the plan creation flow.	Medium	Pass	
FE-	Market Section Navigation	Verifies the navigation link to trading data.	Click the "Market" button in the navigation bar.	The UI navigates to the market analysis section.	Medium	Pass	
FE-CMP-009	Bed Selection	Verifies the initial input method for salt bed count.	Open the plan creation tool.	A range slider or numeric input for salt beds is visible.	High	Pass	
FE-CMP-010	Season Type	Checks the selection UI for the harvest season.	Complete Step 1 and click "Next".	Options for "Fresher season" and "Mid-season" are presented clearly.	High	Pass	

FE-CMP-011	Duration Selection	Validates the time-frame selection interface.	Complete Step 2 and advance.	Selectable buttons or cards for "30 days" and "45 days" are visible.	High	Pass	
FE-CMP-012	Labor Input	Verifies the input field for worker count.	Complete Step 3 and advance.	An input field to enter the number of workers is displayed.	High	Pass	
FE-CMP-013	Projections UI	Checks the visual breakdown of costs and rates.	Complete Step 4 and advance.	A cost breakdown appears showing price/bag (1,550 LKR), carrying (500/bag), and refreshment (350/day).	High	Pass	
FE-CMP-014	Buyer Cards	Validates the presentation of seller recommendations.	Complete Step 5 and advance.	Cards for Colombo Salt Co, Southern Traders, and Lanka Salt Exports appear with prices and star ratings.	High	Pass	
FE-CMP-	Seller Selection UI	Confirms visual feedback when picking a buyer.	Click on a specific seller card.	The selected card is visually highlighted (e.g., green border or checkmark).	Medium	Pass	
FE-	Final Review	Verifies the summary screen before completion.	Complete Step 6 and advance.	A summary of all previous selections is shown with a final "Confirm" button.	High	Pass	
FE-	Step Data Persistence	Checks if data is kept when moving backward.	Navigate to Step 3, enter data, go back to Step 2, then forward to Step 3.	All previously entered data remains populated in the fields.	Medium	Pass	
FE-	Progress Bar UI	Validates the visual tracking of the user's progress.	Move through the steps of the wizard.	A progress bar or step-dots indicator updates at each stage.	Medium	Pass	

FE-CMP-	Math Visibility	Confirms the UI updates totals based on inputs.	Input 3 beds, 2 workers, and 30 days.	The harvest and profit totals update on-screen to reflect the specific calculation.	High	Pass	
FE-CMP-	Price Trends Chart	Verifies the visualization of salt price fluctuations.	Navigate to the Market Analysis section.	A line chart is displayed, correctly plotting salt prices on the Y-axis against time on the X-axis.	Medium	Pass	
FE-CMP-021	Volume Chart	Checks the trade volume data presentation.	Navigate to the Market Analysis section.	A bar chart appears, showing the quantity of salt traded over specific intervals.	Medium	Pass	
FE-CMP-022	Demand Forecast	Validates the predictive demand visualization.	Navigate to the Market Analysis section.	An area chart is visible, showing shaded regions representing future salt demand forecasts.	Medium	Pass	
FE-CRY-001	History Table Layout	Verifies the structural integrity of the production log.	Navigate to the salt production page.	A table is visible with columns for Month, Volume, and Season, along with visible Edit and Delete icons/buttons for each row.	Critical	Pass	
FE-CRY-002	Record Creation UI	Validates the manual entry flow for new production data.	1. Click the "Create" button. 2. Enter Month (YYYY-MM), Volume, and Season. 3. Click "Submit".	A dialog/modal opens for input; upon submission, the modal closes and the new record appears instantly in the table.	Critical	Pass	
FE-CRY-003	Edit Record Dialog	Checks that existing data is correctly loaded for modification.	Click the "Edit" icon on an existing row.	A dialog opens with all fields (Month, Volume, Season) pre-filled with the current values from that row.	High	Pass	

FE-CRY-004	Deletion Confirmation	Verifies the safety check before removing historical data.	Click the "Delete" icon on a row.	A confirmation dialog appears asking the user to verify the deletion before the record is removed from the UI.	High	Fail	Pass
FE-CRY-005	Actual vs. Predicted Chart	Validates the visual comparison of yield performance.	Observe the chart area when data is present.	An area chart displays Actual Production as a solid line and Predicted Production as a dashed line for visual distinction.	Critical	Pass	
FE-CRY-	Month Format Validation	Checks for visual error cues on incorrect date inputs.	Enter an invalid format (e.g., "Jan 2026") into the Month field.	A validation error message appears specifically on the Month field.	Medium	Pass	
FE-CRY-	Volume Constraint UI	Ensures negative numbers are flagged visually.	Enter a negative number in the Volume field.	The UI displays a validation error preventing submission.	Medium	Pass	
FE-CRY-	Season Selection UI	Verifies the available options in the season picker.	Open the Season dropdown menu.	A list of four options appears: spring, summer, autumn, and winter.	Medium	Pass	
FE-CRY-	Empty State UI	Verifies the interface when no history exists.	Access the page with a new account (no records).	An empty state message (e.g., "No records found") is displayed instead of a blank table.	Medium	Pass	
FE-CRY-	Success Feedback	Checks for temporary visual confirmation of actions.	Successfully Create, Edit, or Delete a record.	A success toast notification appears briefly in the corner of the screen.	Medium	Pass	
FE-CRY-011	Error Feedback	Checks for visual alerts during system failures.	Attempt to save a record while the network is disconnected.	An error toast notification appears indicating the action failed.	Medium	Pass	

FE-CRY-012	Daily Parameter Cards	Verifies the display of key crystallization metrics.	Load the production dashboard.	Metric cards are visible for: water temperature, lagoon level, brine levels (OR, IR), bund levels, and channels (East, West).	Critical	Pass	
FE-CRY-013	Weather Metric Display	Checks for the visibility of local climate data.	Observe the weather section of the dashboard.	Rainfall, air temperature, and humidity values are clearly displayed with appropriate units (e.g., °C, %).	High	Pass	
FE-CRY-014	Composed Production Chart	Validates the overlaid visualization of two data types.	Observe the main production chart.	A single chart displays bars (Actual data) and lines (Predicted data) overlaid for direct comparison.	High	Pass	
FE-CRY-015	Date Range Filtering	Tests the responsiveness of charts to time-based filters.	Use the date picker to select a start and end date.	The composed chart and parameter cards update immediately to reflect data from the selected period.	High	Pass	
FE-CRY-016	Forecast Report UI	Checks the presentation of future production estimates.	Click the "Forecast Report" button.	A dialog opens showing a table or list of predicted production values for upcoming days.	High	Fail	Pass
FE-CRY-017	Supervisor Notification UI	Verifies the workflow for alerting management.	Click "Notify Supervisors" and confirm in the dialog.	A confirmation dialog appears; after confirming, a success message indicates supervisors have been notified.	Medium	Pass	
FE-CRY-018	Report Export Options	Validates the UI for downloading production data.	Click the "Download Report" button.	A dialog appears offering "CSV" and "PDF" as selectable export formats.	Medium	Pass	

FE-CRY-	Daily Measurement Entry	Checks the manual data entry form for field teams.	Open the environmental form, fill in current values, and submit.	Upon submission, the dashboard metrics refresh to include the newly entered data.	Critical	Pass	
FE-CRY-020	Historical Data Navigation	Tests UI performance and navigation with high data volume.	Scroll through the historical data list or navigate pages.	The UI remains responsive, using pagination or smooth scrolling to display large datasets without lag.	Medium	Pass	
FE-VIS-035	Period Selector Options	Verifies available reporting intervals.	Open the period selector dropdown.	Dropdown displays: Hourly, Daily, Biweekly, and Monthly.	High	Pass	
FE-VIS-	Hourly Date Picker	Validates UI for single-day reports.	Select "Hourly" from the dropdown.	A single date picker input appears.	High	Pass	
FE-VIS-	Daily Range Picker	Validates UI for multi-day reports.	Select "Daily" from the dropdown.	Both "Start Date" and "End Date" pickers are visible.	High	Pass	
FE-VIS-	Biweekly Range UI	Checks the 14-day window automation.	Select "Biweekly" and choose a date.	The UI automatically highlights or calculates a 14-day range from the selected date.	Medium	Pass	
FE-VIS-	Monthly Picker UI	Validates the month-level selection interface.	Select "Monthly" from the dropdown.	A Month/Year selector (rather than specific days) is shown.	Medium	Pass	
FE-VIS-	Report Generation (Hourly)	Confirms data population for hourly views.	Select Hourly + date and click "Generate".	Summary cards and the breakdown table populate with data for that specific day.	Critical	Pass	
FE-VIS-	Report Generation (Daily)	Confirms data population for range views.	Select Daily + range and click "Generate".	Summary cards and table populate with data across the selected dates.	Critical	Pass	

FE-VIS-042	Summary Statistic Cards	Verifies the display of key performance indicators.	Generate any report.	Visual cards show: Total detections, Pure, Impure, Unwanted counts, Avg purity, and Detections/hour.	High	Pass	
FE-VIS-	Hourly Table Layout	Checks the structural integrity of hourly logs.	Generate an Hourly report.	Table shows: Hour, Detections, Pure, Impure, Unwanted, and Avg Purity per hour.	High	Pass	
FE-VIS-	Daily Table Layout	Checks the structural integrity of daily logs.	Generate a Daily report.	Table shows: Date, Detections, Pure, Impure, Unwanted, and Avg Purity per day.	High	Pass	
FE-VIS-	CSV Export Flow	Validates the raw data export functionality.	Click the "Export CSV" button.	A CSV file downloads containing all data from the generated report table.	High	Fail	Pass
FE-VIS-	PDF Export Flow	Validates the formatted report export.	Click the "Export PDF" button.	A PDF file downloads containing the summary cards and the data table.	High	Fail	Pass
FE-VIS-	Empty State UI	Checks the interface for periods with no data.	Generate a report for a date with no recorded detections.	The UI displays a "No data available" message.	Medium	Pass	
FE-VIS-	Generation Loading State	Verifies visual feedback during processing.	Click "Generate" and observe the button.	The button displays a loading animation and becomes disabled until the report is ready.	Medium	Pass	
FE-VIS-	Error Alert UI	Validates user feedback for system failures.	Attempt to generate a report during a network failure.	An error alert or banner appears with a descriptive message.	High	Pass	
FE-ADM-	Paginated User List	Verifies the core data table layout and row count.	Navigate to the users page.	A table displays 10 users per page with columns for: ID, Email, Phone, Name, Role, and Status.	Critical	Pass	

FE-	Next Page Navigation	Validates the ability to view subsequent user records.	Click the "Next" pagination button.	The table refreshes to show the next set of users (Page 2).	High	Fail	Pass
FE-	Previous Page Navigation	Validates returning to previous record sets.	On page 2, click the "Previous" button.	The table refreshes to show the previous set of users (Page 1).	High	Pass	
FE-ADM-007	Page Counter UI	Checks the visual indicator for current position in the list.	Observe the pagination area when multiple pages exist.	A label displays "Page X of Y" (e.g., Page 1 of 5).	Medium	Pass	
FE-ADM-	Status Badge UI	Verifies the visual representation of user states.	View the status column in any user row.	Distinct badges are visible for: Onboarded, Verified, Subscribed, and Trial Active.	High	Pass	
FE-ADM-	Create User Dialog	Checks the layout of the manual user addition form.	Click the "Create User" button.	A dialog opens containing fields for: Email, Password, Name, and a Role dropdown.	Critical	Pass	
FE-ADM-	Valid User Creation	Confirms UI behavior after successfully adding a user.	Fill all fields in the "Create User" dialog and click "Submit".	The dialog closes automatically and the user list refreshes to include the new entry.	Critical	Pass	
FE-	Required Field Validation	Tests visual error handling in the creation form.	Leave the Email field empty and attempt to submit the form.	A validation error message appears specifically on the Email field.	High	Pass	
FE-ADM-012	Edit User Dialog UI	Validates that existing data loads correctly for updates.	Click "Edit" on a specific user row.	A dialog opens pre-filled with the user's Name, and toggles for Onboarding, Plan, Subscription, and Verification.	High	Pass	
FE-	Save User Changes	Confirms UI behavior after a successful edit.	Modify the Name in the edit dialog and click "Save".	The dialog closes and the user row in the table reflects the updated name.	High	Pass	

FE-	Delete Confirmation UI	Verifies the safety check before account removal.	Click "Delete" on a user row.	A confirmation dialog appears asking: "Are you sure?"	High	Pass	
FE-ADM-	Confirm User Deletion	Confirms UI behavior after account removal.	Click the confirm button in the delete dialog.	The dialog closes and the specific user is immediately removed from the table view.	High	Pass	
FE-	Verify User UI Action	Validates the manual verification trigger.	Click the "Verify" button on an unverified user row.	The user's status badge visually updates to show "Verified" status.	High	Pass	
FE-	User Search Bar	Tests the UI's ability to filter records.	Type a name or email into the search input field.	The user list updates in real-time to show only matching records.	Medium	Pass	
FE-ADM-018	Role Selection Options	Verifies the complete list of available system roles.	Open the "Role" dropdown in the create or edit form.	Options include: SUPERADMIN, ADMIN, LANDOWNER, DISTRIBUTOR, SELLER, LABORATORY, SALTSOCIETY, and VIEWER.	High	Pass	
FE-ADM-019	Plan Card Layout	Verifies the summary view for all configured subscription tiers.	Navigate to the plans page.	Cards display: Name, Key, Level, Monthly/Annual prices, Features, and Active status.	High	Pass	
FE-ADM-020	Create Plan Dialog	Checks the configuration form for new service tiers.	Click the "Create Plan" button.	A dialog opens with fields for: Key, Name, Level, Monthly/Annual prices, Features (comma-separated), and Duration.	High	Pass	
FE-ADM-	Submit New Plan	Confirms UI updates after adding a new subscription tier.	Fill valid data in the dialog and click "Submit".	The dialog closes and a new plan card appears instantly on the dashboard.	High	Pass	

FE-ADM-	Edit Plan Dialog UI	Validates that current settings load correctly for modification.	Click the "Edit" button on any plan card.	A dialog opens with all fields pre-populated with that plan's current values.	High	Pass	
FE-ADM-023	Save Plan Updates	Confirms visual consistency after modifying plan details.	Modify a price or feature in the dialog and click "Save".	The dialog closes and the respective plan card reflects the new changes.	High	Pass	
FE-ADM-	Plan Deactivation UI	Verifies the process for retiring a subscription tier.	Click "Deactivate" on a non-free plan card.	A confirmation dialog appears; upon confirming, the card status updates to "Inactive".	High	Pass	
FE-ADM-	Free Plan Protection	Ensures the mandatory "Free" tier cannot be disabled.	Locate the card with the "free" key.	The "Deactivate" button is either hidden or visually disabled (grayed out).	Medium	Pass	
FE-	Feature Badge Display	Checks the UI formatting for plan capabilities.	Observe the feature list on any plan card.	Each individual feature key is rendered as a distinct visual badge.	Medium	Pass	
FE-ADM-	Plan Status Badges	Validates clear visual indicators for plan availability.	View the status section of any plan card.	A green "Active" badge or a red "Inactive" badge is clearly visible.	Medium	Pass	
FE-ADM-028	Payment List Display	Verifies the core data columns for financial tracking.	Navigate to the payments page.	A table is visible containing: Order ID, User ID, Plan, Billing Cycle, Amount, Currency, Status, and Date.	High	Pass	
FE-ADM-	Financial Pagination	Validates the layout for high-volume transaction sets.	Observe the table when more than 20 payments exist.	Pagination controls are visible; the table displays exactly 20 records per page.	Medium	Pass	

FE-ADM-030	Transaction Status UI	Checks the color-coded feedback for payment states.	View the status column across multiple rows.	Badges appear as: Success (green), Pending (gray/secondary), Failed (red), and Refunded (outline).	High	Pass	
FE-ADM-	Currency Formatting	Ensures financial values are localized and readable.	View the "Amount" column in any payment row.	The value is formatted with the currency code (e.g., "1,500 LKR").	Medium	Pass	
FE-ADM-032	Subscription List	Verifies the display of active service agreements.	Navigate to the subscriptions page (/superadmin/subscriptions).	A list or table is displayed showing all active subscriptions, including the User, assigned Plan, and Start/End Dates.	Medium	Pass	
FE-ADM-033	Audit Log Table	Verifies the comprehensive data structure of system logs.	Navigate to the audit logs page (/superadmin/audit-logs).	A table is visible with columns: Service, Action, User ID, Resource, Status, Method, Path, and Timestamp.	High	Pass	
FE-ADM-	Service Filtering	Tests the UI's ability to isolate logs by specific modules.	Open the service filter and select a specific service (e.g., "Vision").	The table refreshes to show only logs associated with the selected service.	Medium	Pass	
FE-	Action Filtering	Validates log isolation by operation type.	Open the action filter and select a type (e.g., "DELETE").	The UI displays only log entries that match the selected action type.	Medium	Pass	
FE-ADM-036	Date Range Filter	Checks time-based log auditing.	Use the date pickers to set a start and end date.	The logs are filtered to show only actions that occurred within the chosen timeframe.	Medium	Fail	
FE-ADM-	Log Pagination	Ensures navigation through high-volume event data.	Observe the table when a large set of logs is present.	Paginated navigation controls appear, allowing the user to move between sets of logs.	Medium	Pass	

FE-ADM-038	Detailed Log View	Checks the expanded visualization of a single log event.	Click on a specific log entry in the table.	A side panel or modal opens showing the full JSON or detailed breakdown of that specific log.	Medium	Pass	
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8. Observability & Runbook

- **Monitoring:** CloudWatch Metrics track Approximate Number Of Messages Visible for SQS and CPU/Memory for ECS.
- **Logging:** Structured JSON logs are aggregated for correlation across microservices.
- **Recovery:** If SQS backlog grows beyond a threshold, an Auto-Scaling policy triggers additional worker tasks.